

Theories of motivation: A comprehensive analysis of human behavior drivers

Din Bandhu^a, M. Murali Mohan^b, Noel Anurag Prashanth Nittala^c, Pravin Jadhav^d, Alok Bhadauria^{a,*}, Kuldeep K. Saxena^e

^a Department of Mechanical and Industrial Engineering, Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal 576104, Karnataka, India

^b Department of Mechanical Engineering, G. Pulla Reddy Engineering College (Autonomous), Kurnool 518007, Andhra Pradesh, India

^c Department of Sciences, Indian Institute of Information Technology Design and Manufacturing (IIITDM), Kurnool 518008, Andhra Pradesh, India

^d Department of Economics, Institute of Infrastructure, Technology, Research and Management (IITRAM), Ahmedabad 380026, Gujarat, India

^e Division of Research and Development, Lovely Professional University, Phagwara, India

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ABSTRACT

This paper explores theories of motivation, including instinct theory, arousal theory, incentive theory, intrinsic theory, extrinsic theory, the ARCS model, self-determination theory, expectancy-value theory, and goal-orientation theory. Each theory is described in detail, along with its key concepts, assumptions, and implications for behavior. Intrinsic theory suggests that individuals are motivated by internal factors like enjoyment and satisfaction, while extrinsic theory suggests that external factors like rewards and social pressure drive behavior. Arousal theory says that to feel motivated, people try to keep an optimal level of activation or excitement. Incentive theory suggests that behavior is driven by the promise of rewards or the threat of punishment. The ARCS model, designed to motivate learners, incorporates elements of attention, relevance, confidence, and satisfaction. Self-determination theory proposes that individuals are motivated by their needs for autonomy, competence, and relatedness. The expectation-value theory suggests that behavior is influenced by individuals' beliefs about their ability to succeed and the value they place on the task. The goal-orientation theory suggests that individuals have different goals for engaging in a behavior. By understanding these different theories of motivation, educators, coaches, managers, and individuals may analyze what drives behavior and how to harness it to achieve their goals. In essence, a nuanced comprehension of these diverse motivation theories equips individuals across varied domains with a strategic toolkit to navigate the complex landscape of human behavior, fostering a more profound understanding of what propels actions and how to channel these insights toward the attainment of overarching goals.

1. Introduction

Human behavior is influenced by a complex interplay of numerous factors. These include biological determinants, psychological factors, social and cultural elements, economic and environmental factors, as well as external stressors that play a significant role in shaping how individuals interact and respond to their environment. This paper focuses on understanding motivation, a psychological factor that impacts human behavior. Motivation represents the yearning to accomplish a task, paired with the enthusiasm and determination to see it through. It acts as the driving force that propels an individual to take proactive steps

and reach their goals. The term “intrinsic” refers to motivation that originates internally in the form of a person's interests and goals, while “extrinsic” refers to motivation that is prompted by external variables like incentives and penalties (Amaro et al., 2021; Harris & McDade, 2018; Sharma & Gupta, 2022).

Motivation can be positive, which means it inspires a person to take positive actions and make positive changes, or negative, which means it leads a person to take negative actions or make negative changes (Eccles & Wigfield, 2002; Gollwitzer & Oettingen, 2001; Schiefele, 1991). There is no standard definition for motivation. Nevertheless, it has been defined by several scholars in their respective perspectives. According to

* Corresponding author.

E-mail address: alok.bhadauria@manipal.edu (A. Bhadauria).

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Dwight D. Eisenhower (Low, 2013), “Motivation is the art of getting people to do what you want them to do because they want to do it”, which explains that motivation may be used as an agent to getting one's tasks accomplished by encouraging people without them knowing. Zig Ziglar defined motivation as fuel, necessary to keep the human engine running (Mcalister, 2018), proving that motivation is essential to keep human life going. John Adir claimed that “Motivation is the driving force within an individual that impels them to pursue their goals” (Schiefele, 1991) in contrast to Dwight who said it can be used to get the work accomplished by others. Similarly, Tony Robbins defined motivation as the catalyst that sets the engine of success in motion (Robbins, 2011), which again reiterates the fact that motivation is crucial for an individual to succeed in life.

Numerous hypotheses have been put out to try and make sense of what drives humans. Thus, it is difficult to determine who gave the first definition of motivation in education, as the concept of motivation has been studied and discussed by educators, psychologists, and philosophers for centuries. While every theory provides insight into a certain facet of motivation, they all inevitably leave out others. Ambiguity arises from the wide variety of ideas because they all share some common ground but also disagree on key points, and many of them use a jargon that is unique to them (Cook & Artino, 2016a; Murphy & Alexander, 2000; Petri, 2012). Despite the unsettling nature of this, it is important to remember that each theory offers a fresh viewpoint that has the potential to yield novel insights and different implications for learning and research activities.

In the past, analyses of motivation have either been centered on the theories' practical applications, provided very cursory overviews, or concentrated on a single theory (Artino, 2012; Kusurkar et al., 2011; Mann, 1999; Teunissen & Bok, 2013). Further contemplation is needed to enhance this field of study, and a review that discusses and analyzes numerous theories would inspire a more nuanced knowledge of motivational principles. In this regard, this article condenses nine theories on learning motivation and illustrates key parallels and differences between them to set the stage for future study. The purpose of this article is not to provide a thorough analysis of recent research findings but rather to make the theoretical underpinnings of motivation comprehensible to learners in any discipline to acquire knowledge. Therefore, this article gives a synopsis of motivational theories that influence people's actions in learning environments. However, further readings are suggested for individuals interested in delving deeper into these theories.

Furthermore, any discussion on motivation is incomplete without mentioning the earliest and most influential theories of motivation in education developed by Abraham Maslow in the 1940s, who proposed a hierarchy of needs that outlined the different types of motivation that drive human behavior. According to Maslow, people are motivated by physiological needs, safety needs, belongingness and love needs, esteem needs, and self-actualization needs, with each level of needs building upon the previous one. This theory suggests that human needs are organized into a hierarchy, with the most basic needs at the bottom and the most complex needs at the top. Maslow believed that humans must satisfy their lower-level needs before they can move on to satisfying their higher-level needs. Maslow's hierarchy of needs is a widely recognized and influential theory of human motivation. It is often discussed in psychology, business, and education because it provides a framework for understanding what motivates people. Maslow's hierarchy of needs provides a model for understanding what drives people to behave in the way they do. This understanding can be helpful in a variety of contexts, such as designing workplaces that are motivating for employees or developing educational programs that are engaging for learners. Since then, several academicians explored the term motivation and molded its definition according to their needs and situations (Collins, 2009; Cook & Artino, 2016b; Dinbandhu & Jadhav, 2020; Schunk & Zimmerman, 2012; Tracy, 2018). In this context, Nicholls et al. (Nicholls et al., 1989) defined motivation as the processes that arouse, direct, and maintain behavior toward a specific goal. Weiner

(Weiner, 1992) also came up with a definition of motivation as a psychological construct that refers to the forces that initiate, energize, direct, and sustain behavior, often toward some specific goal or reward. Zeidner et al. (Zeidner et al., 2000) defined motivation as an individual's internal drive or desire to engage in a particular behavior or activity. It is the psychological process that initiates, guides and maintains goal-oriented actions. Schunk (Schunk, 2004) saw motivation as the process by which goals are activated and sustained, and it involves the arousal, direction, and persistence of behavior.” Also, Guerin et al. (Guerin et al., 2011) defined motivation as the psychological process that gives behavior purpose and direction. It is what causes us to act, whether it is getting a glass of water to reduce thirst or reading a book to gain knowledge.”

From the aforementioned discussion, it is clear that motivation drives learners to engage in learning and pursue their educational goals. Without motivation, learners may struggle to find the energy and determination to complete assignments, prepare for exams, and achieve success in their studies (Haque et al., n.d.; Schürmann et al., 2021; Tohidi & Jabbari, 2012). Motivation can come from a variety of sources, including personal interests, future career aspirations, and a desire to make a positive impact on the world. In the academic setting, many challenges and obstacles can hinder motivation. These may include the demands of coursework, the pressure to succeed, and the fear of failure. Additionally, learners may face external factors such as financial challenges, family responsibilities, and other personal commitments that can make it difficult to prioritize their education. A pictorial representation of the motivation to learn is shown in Fig. 1 (Shkoler & Kimura, 2020; Steinmayr et al., 2019; Vo et al., 2022).

Fig. 1 depicts the internal and external factors that drive the motivation to learn. A healthy balance between both these factors is essential for learning to take place effectively. Learners need to find ways to stay motivated and inspired in their academic pursuits. This may involve setting clear and achievable goals, seeking support and guidance from mentors and peers, and finding ways to stay engaged and interested in the material being studied. Building a strong support network and developing effective study habits and strategies can also be instrumental in helping learners maintain their motivation and achieve success in their academic endeavors. Overall, learners need to find ways to stay motivated and focused in order to reach their full potential and achieve their goals (Dweck et al., 2014; Pintrich & De Groot, 2003; Sethi & Scales, 2020; Van Vu et al., 2022).

From existing literature, it has been observed that motivation is a psychological drive that energizes and directs behavior toward a goal. It is a complex phenomenon that is influenced by a variety of factors, including biology, psychology, social and cultural factors, economics, the environment, and external stressors. In the academic setting, motivation is essential for students to succeed. Without motivation, learners may struggle to find the energy and determination to complete assignments, prepare for exams, and achieve their educational goals. This paper reviews nine theories of learning motivation: instinct theory, arousal theory, incentive theory, intrinsic theory, extrinsic theory, the ARCS model, self-determination theory, expectancy-value theory, and goal-orientation theory. The focus will be on understanding the key concepts, assumptions, and implications that these theories might have on human behavior.

2. Learning motivation theories

Motivation is a psychological construct that drives individuals to engage in certain behaviors or activities. In the context of learning, motivation refers to the factors that drive an individual to engage in the process of acquiring knowledge and skills (Willy, n.d.; Hilgard, 2004; Svinicki & Vogler, 2012). Several motivation theories can influence learning.

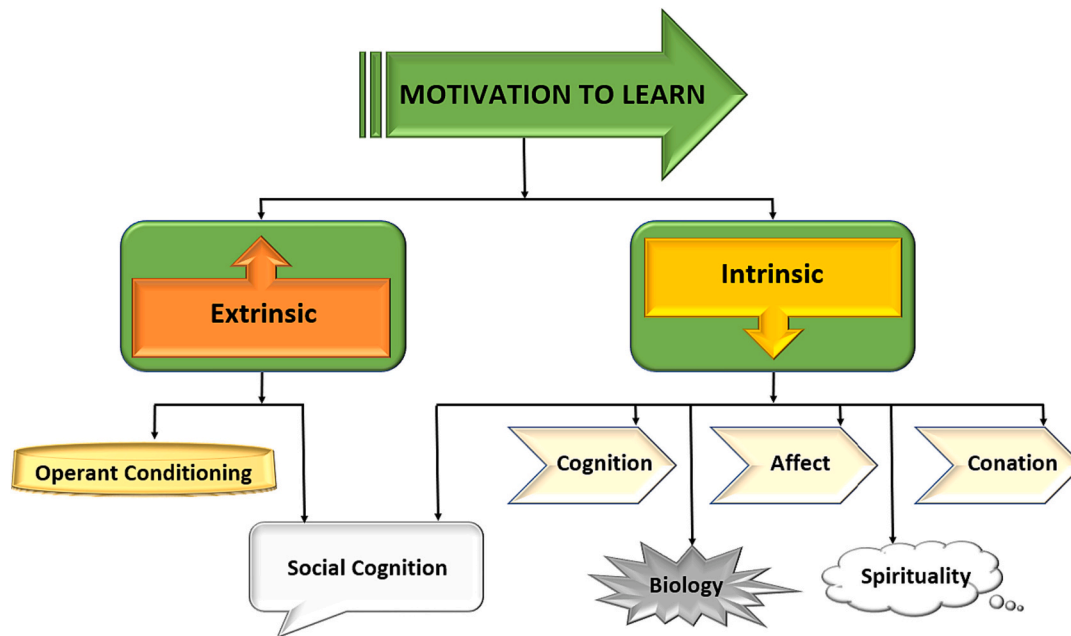


Fig. 1. A social framework of motivation to learn.

3. Instinct theory of motivation

Instinct theory, which has been considered as a root for all motivation theories, was very popular in the early 20th century. The term ‘instinct’ was coined by Charles Darwin and George Romanes in the late 19th century to explain the behavior of animals. It is a psychological theory that suggests that certain behaviors are innate or hard-wired into individuals and that these behaviors are triggered by specific stimuli or cues in the environment. The theory suggests that certain behaviors are not learned through experience but are instead pre-programmed into the individual's biology. It is important to note that this theory has been falling out of favor lately in several recent researches, which suggest that most behaviors are the result of a combination of genetic, environmental, and cultural influences. Critics of the instinct theory argue that it is difficult to define an instinct clearly and objectively and that many behaviors that have been labeled as “instinctive” are the result of complex interactions between genes and the environment. Furthermore, many behaviors that appear to be instinctive in one species are not present or are much less prominent in closely related species, which suggests that they are not solely determined by genetics (Burnham, 1972; Gillespie, 1971; Harlow, 2004; Loewald, 1972; Loewald, 1978; Oppenheimer, 1958). Instinct theory suggests that some behaviors are innate and not learned. In the context of learning, it implies that certain behaviors may be driven by instinctual responses rather than acquired knowledge or conditioning.

4. Arousal theory of motivation

Arousal theory also comes under the category of prominent theories of motivation. It is a psychological theory introduced by Robert Yerkes and John Dillingham Dodson in 1908. It suggests that there is an optimal level of physiological or psychological arousal that is necessary for optimal performance on a given task. According to this theory, when an individual is too aroused (i.e., over-aroused), performance may suffer, while when the individual is not aroused enough (i.e., under-aroused), performance may also suffer. The theory posits that there is an “optimal” level of arousal that is associated with the best performance. The well known version of this theory is the Yerkes-Dodson law, which states that as arousal level increases, performance will also increase up to a point, after which it will decrease. Arousal can be caused by a wide

range of factors, including physiological processes such as the release of hormones, psychological factors such as stress or motivation, and environmental factors such as noise or temperature. It's important to note that the theory is based on an inverted-U shape relationship between performance and arousal, meaning that there is a point where high arousal levels lead to diminishing returns and that optimal performance will be achieved at moderate levels of arousal. Arousal theories have been applied in various fields such as sports, education, and cognitive psychology. It has been used to explain the effects of stress, anxiety, and other emotional states on human performance (Kerr, 1985; Kerr, 2021; Reizenzein, 1994; Reizenzein, 2017). Arousal theory posits an optimal level of stimulation for effective learning. In learning an appropriate balance of excitement and calmness can enhance the ability to acquire and retain information.

5. Incentive theory of motivation

The *incentive theory*, which was first proposed by a behaviorist, Clark Hull in 1943. Hull proposed that behavior is a function of drive, which is a physiological state that creates a need, and incentive, which is the positive or negative values of a particular situation. This theory explains how behavior is influenced by the presence or absence of incentives, or rewards. The theory suggests that the magnitude and proximity of an incentive can influence behavior by increasing the perceived value of a particular action or outcome, which in turn increases the motivation to engage in that behavior. It further proposes that behavior is driven by an organism's attempt to maximize pleasure and minimize pain. Incentives can be positive or negative, they can be external, such as money, or internal, such as hunger or thirst. Incentive theory has been used to explain a wide range of behaviors, such as drug addiction, consumer choice, and work motivation. It has also been used to explain how rewards and punishments can influence behavior in educational and work settings. Recent research has suggested that incentives are just one of many factors that can influence behavior and that the effectiveness of incentives may depend on the specific context and individual characteristics. It's worth mentioning that there are some criticisms of the theory, some argue that the theory oversimplifies the complexity of human motivation and can be reductionist in nature (Bosha et al., 2017; Ellingsen & Johannesson, 2008; Hattie et al., 2020; Killeen, 1982; Vi & Thuy, 2020). Incentive theory emphasizes the role of rewards and

motivations in influencing behavior. In learning, it suggests that offering incentives, such as rewards or positive outcomes, can stimulate engagement and enhance the learning process.

In its most basic form, the concept of motivation may be broken down into three distinct categories: intrinsic motivation, extrinsic motivation, and amotivation (Gopalan et al., 2017). In addition, there are a number of additional theories that have the potential to be put into practice, most notably in the field of education. These theories are the attribution theory (Weiner, 1972), self-determination theory (Gagné, & E.D.-J. of O. behavior, undefined, 2005), the ARCS model (Keller, 2010), the social cognitive theory (Bandura, 1991), and the expectancy theory (Wabba & House, 1974). These educational theories are capable of existing on their own and making a contribution to the result of the learning process without requiring the support of any other educational theories. These theories have been explained in the following section:

6. Intrinsic theory of motivation

The intrinsic theory of motivation is a psychological theory that suggests that individuals engage in certain behaviors or activities because they find them personally interesting or satisfying, rather than because of the potential external rewards or pressures. Intrinsic motivation is considered to be self-driven and is characterized by feelings of enjoyment, interest, and satisfaction in the activity itself (Ryan & Deci, 2020).

Edward Deci originally put out the concept of intrinsic motivation in the 1970s. According to this theory, people's motivation levels are at their highest when they can fulfill their basic psychological requirements. According to Deci, human beings have a desire for autonomy, competence, and relatedness. The need for autonomy relates to the desire to direct one's own life and make decisions, the need for competence to the conviction that one possesses the skills necessary for success, and the need for relatedness to the satisfaction that comes from having meaningful relationships with others (Deci, 1972).

Deci's theory of intrinsic motivation has been employed in a variety of fields, including education, sport, and the workplace, and has been used to explain why individuals engage in activities such as learning, creativity, and volunteering. Research suggests that activities that meet these three basic needs, tend to be more intrinsically motivating, than those that do not, and also tend to lead to better performance, satisfaction, and well-being (Deci et al., 2001; E.D.-O. behavior and human performance, undefined, 1976).

Intrinsic motivation is considered to be a more sustainable and long-term form of motivation, as it is driven by personal interests and values, rather than external rewards. Nonetheless, this doesn't mean that the external rewards are not important, they can also play a role in the process of motivation, depending on the context and the individual (Deci & Ryan, 2013; Ryan & Deci, 2020). The intrinsic theory of motivation suggests that people are naturally driven to learn when they find the subject matter engaging and personally meaningful. In learning, this means that educators can enhance motivation by making the content relevant and tapping into students' inherent curiosity and interest.

7. Extrinsic theory of motivation

Extrinsic theory suggests that individuals engage in certain behaviors or activities because of the potential external rewards or pressures, rather than because they find them personally interesting or satisfying. Extrinsic motivation is considered to be externally driven and is characterized by the presence of external factors such as grades, money, or praise that serve as incentives to engage in a behavior or activity. Extrinsic motivation can take many forms, such as rewards, like receiving a prize for completing a task, or punishment, such as being given a penalty for not completing a task. The theory suggests that individuals engage in activities in order to earn rewards or avoid punishments (Deci et al., 1991; Ryan & Deci, 2000; Ryan & Deci, 2020).

Extrinsic motivation can also be classified as *controlled motivation* and *integrated motivation*. *Controlled motivation* refers to behavior that is performed because of external rewards or to avoid punishment. *Integrated motivation* refers to behavior that is performed to gain an external reward but also aligns with an individual's interests and values (Ryan & Deci, 2020).

Although the theory of extrinsic motivation was proposed by several authors, B.F. Skinner in the 1950s was the pioneer who emphasized that behavior is controlled by its consequences and that we tend to repeat behaviors that lead to positive outcomes and avoid behaviors that lead to negative outcomes (Jones & Skinner, 1939). Extrinsic motivation has been applied in a wide range of fields, such as education, sports, and the workplace, and has been used to explain why individuals engage in activities such as studying, working, and volunteering. Studies suggest that while extrinsic rewards can be effective in increasing motivation and performance in the short term, they may have negative effects on motivation and performance in the long term, as they can reduce the sense of autonomy and interest in the activity (Duggal et al., 2021; Razali et al., 2020; Sun & Hsieh, 2018). Extrinsic motivation theory posits that external rewards or punishments can drive learning by offering incentives for desired behaviors. In education, this implies that students may be motivated to learn when they are offered tangible rewards or avoid negative consequences, but the challenge lies in maintaining long-term engagement and fostering genuine interest in the subject matter.

8. Intrinsic theory of motivation as contrasting to extrinsic theory of motivation

Intrinsic motivation theory and extrinsic motivation theory represent two distinct approaches to understanding the driving forces behind human behavior. Intrinsic motivation centers on the inherent satisfaction and enjoyment derived from engaging in an activity for its own sake, without the need for external rewards. This theory posits that individuals are naturally inclined to pursue tasks that align with their interests and values, fostering a genuine passion for learning. On the other hand, extrinsic motivation theory focuses on external factors, such as rewards or punishments, as catalysts for behavior. It suggests that individuals are motivated by external incentives, like grades or monetary rewards, to achieve desired outcomes. While both intrinsic and extrinsic motivations play roles in shaping human behavior, the former is often associated with sustained interest, creativity, and a deeper level of engagement, whereas the latter may lead to more temporary compliance driven by external factors. Striking a balance between intrinsic and extrinsic motivators is crucial in cultivating a holistic and enduring approach to motivation in various contexts, including education.

9. Attribution theory of motivation

The attribution theory of motivation is a psychological discipline that examines the effects of people's beliefs about the nature of causation on their motivation. Attribution theory suggests that people tend to attribute the cause of an event to either internal or external factors and that these attributions can influence their motivation to engage in similar behaviors in the future. To attribute an event to internal variables, like one's own abilities or the amount of work put in, is called an internal attribution, for example, when someone excels in a sports competition, they may attribute their success to their own skills, dedication, or hard work. Similarly, attributing an event to external elements, like chance or the actions of others, is called an external attribution for example, if a learner performs poorly on a test, they might attribute it to the difficulty of the questions rather than their own lack of preparation (Graham, 1991; Graham, 2020; Weiner, 2010).

Attribution theory was developed by Fritz Heider in the 1950s and later expanded upon by Harold Kelley and others (Muschetto & Siegel,

2021). It proposes that people have a fundamental need to understand the causes of events and that they use a variety of cues to make judgments about the causes of events. The theory also suggests that these judgments can influence motivation by affecting an individual's perceptions of their ability, control, and self-esteem. For example, if an individual attributes success to their ability, they are more likely to feel confident in their ability to achieve similar success in the future, and thus more motivated to engage in similar behaviors. On the other hand, if an individual attributes a failure to external factors, they may feel less responsible for the outcome, and therefore less motivated to engage in similar behaviors in the future (Moring, 2022; Spitzberg & Manusov, 2021; Suhartini et al., 2019).

Fig. 2. presents Weiner's point of view concerning the attribution theory. An incident triggers the process and depending on the outcome, good or negative feelings (pleasure or irritation) may be experienced immediately. However, the curious "nave scientist" who strives to uncover a causal interpretation is generally roused by unanticipated, unfavorable, or seemingly significant events. Each person will "hypothesize" a cause based on their unique perspective on the situation, which will vary across 3 dimensions: locus, stability, and controllability. Stability influences one's confidence in their own future progress. Emotional reactions (which represent perceptual significance) are affected by locus, controllability, and stability, all of which in turn regulate subsequent behaviors (Weiner, 1972).

Attribution theory has been applied in a wide range of fields,

including education, sport, and the workplace, and has been used to explain how factors such as feedback, rewards, and attributions can influence motivation. The theory is also closely related to self-esteem and self-concept and it helps to understand how people evaluate themselves and how that evaluation affects their motivation (Cook & Artino, 2016a; Motevalli et al., 2020; Price, 2021). Individual differences significantly impact motivation in terms of attribution, as people may attribute success or failure to factors like ability, effort, or external circumstances based on their unique personalities, experiences, and beliefs. These varying attributions can shape individuals' motivation levels and affect their future behaviors and learning outcomes. Attribution theory focuses on how individuals attribute their successes and failures, impacting their motivation and future learning behavior. In learning, when learners attribute their successes to their efforts or strategies and view failures as opportunities for growth, they are more likely to remain motivated and persist in their educational pursuits (Harvey & Martinko, n.d.; Baumeister et al., 1990; Saleh & Sturm, 2016).

10. Expectancy theory of motivation

According to the expectancy theory of motivation, people are prompted to act, if they think doing so will bring about positive results. The theory suggests that expectancy, instrumentality, and valence are the three most important components of expectancy motivation. The expectancy is that hard work will provide positive results. The idea that

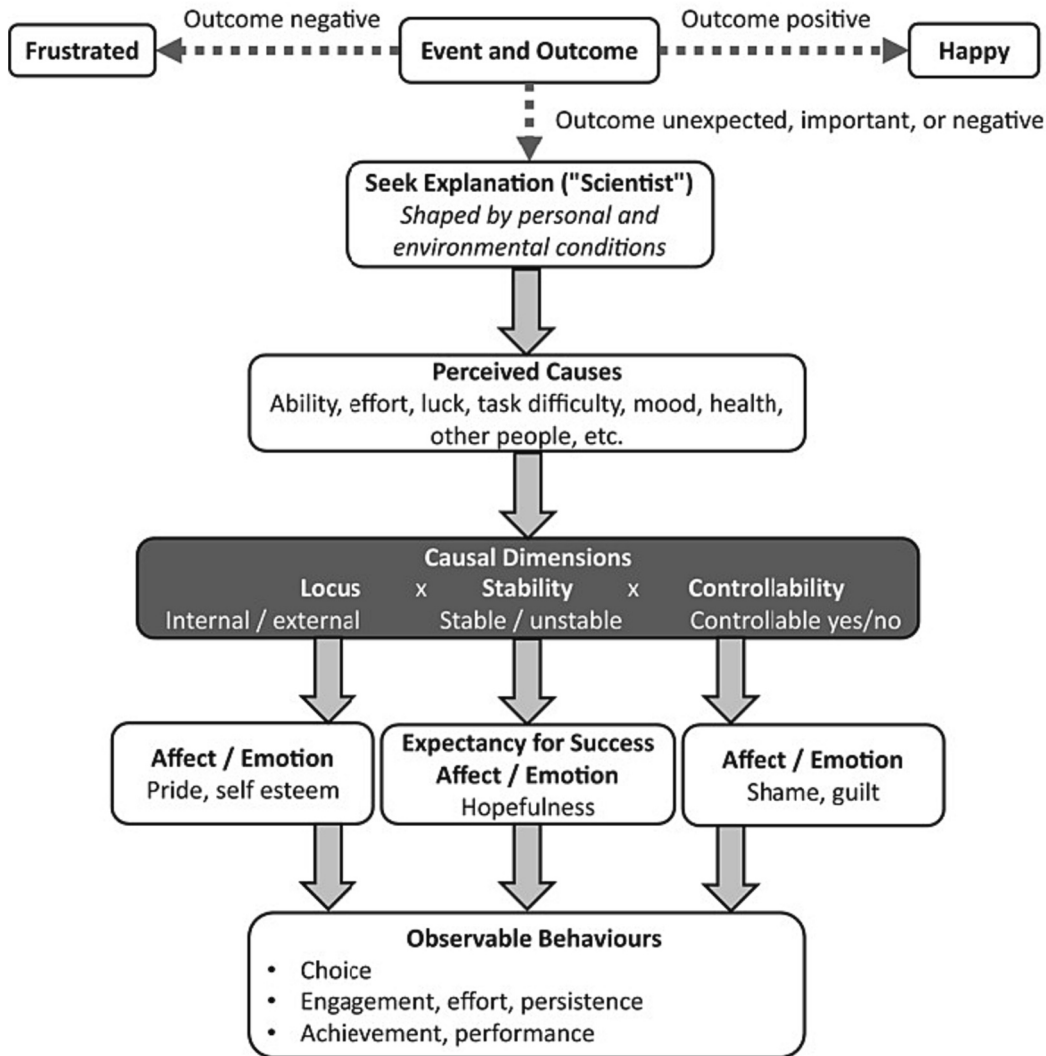


Fig. 2. Weiner's attribution theory (Weiner, 1972).

one's efforts should be rewarded with the best possible results is known as "instrumentality." The term "valence" is used to describe the significance of an outcome to a person (Chopra, 2019; Fudge et al., 1999; Isaac et al., 2013).

Expectancy theory was developed by Victor Vroom in the 1960s, and it suggests that motivation is a function of expectancy, instrumentality, and valence (Parijat & Bagga, 2014). Fig. 3 depicts the framework of this theory.

According to this theory, an individual will be motivated to engage in a behavior if they believe that: i). their effort will lead to high performance (expectancy), ii). high performance will lead to the desired outcomes (instrumentality), and iii). the desired outcomes are valuable or important to them (valence). The theory also proposes that people may be motivated to engage in different behaviors depending on the strength of their expectancies, the perceived instrumentality, and the valence of the outcomes. Expectancy theory has been employed in a variety of fields, including education, sport, and the workplace, and has been used to explain how factors such as goal-setting, feedback, and rewards can influence motivation (Estes & Polnick, 2012; Geiger & Cooper, 1996).

It's important to mention that the Expectancy theory is one of the most influential theories of motivation in the workplace and it has been widely used in management, human resources, and organizational behavior. Expectancy theory has been found to be useful in understanding how people make decisions regarding their efforts and how they evaluate the outcomes of their efforts (Dejong et al., 2001; Sneed & Harrell, 1994). Expectancy theory in the context of learning posits that learners are motivated when they believe that their efforts will lead to successful outcomes and that these outcomes are personally valuable. Learners who expect their hard work will result in positive educational achievements are more likely to remain engaged and committed to their learning goals.

11. Expectancy-value theory of motivation

The expectancy-value theory is a psychological theory of motivation that advocates that people are driven to engage in a behavior or activity based on their beliefs about the likelihood (expectancy) of success and the value (importance or desirability) of the outcome. The expectancy-value theory is rooted in the idea that individuals make choices based on their perceived ability to successfully complete a task and the

perceived value of the outcome (Shang et al., 2022; Wigfield, 1994).

The theory suggests that motivation is determined by the product of expectancy (E) and value (V), or $\text{motivation} = E \times V$. Expectancy represents an individual's belief that they can successfully complete a task, while value represents the importance or desirability of the outcome. The theory was first proposed by Martin Fishbein and Icek Ajzen in the 1970s and it has been applied to various fields such as education, sport, and the workplace (Fishbein & Ajzen, 1973; Fishbein & Ajzen, 2005).

Using a streamlined form of Wigfield and Eccles' theory, Fig. 4 depicts the expectancy-value theory. It does not cover every element of their theory and muddles several small but possibly significant distinctions. Motivational theories, which are driven by social factors that are seen and understood by cognitive processes in the learner, impact the important notions of task value and anticipation of success. The expectancy-value theory has been used to explain why individuals choose to engage in certain behaviors or activities, and how motivation can be increased or decreased by changing the perceived expectancy and value of the outcome. Research suggests that when individuals perceive that their chances of success are high and that the outcome is valuable, they will be highly motivated to engage in the behavior or activity. The expectancy-value theory has been used to design instruction, to improve motivation and performance. For example, teachers can increase learners' motivation by making clear the learning objectives, providing appropriate feedback, and fostering a sense of self-efficacy. It's worth noting that the Expectancy-value theory is not the only theory that explains motivation, and other factors such as emotional states, cognitive processes, and social context should also be taken into account (Otsuka & Smith, 2005; Rachmatullah et al., 2021; Wigfield & Cambria, 2010). Expectancy-value theory suggests that learner motivation is influenced by their expectations of success and the perceived value of a task. In learning, when learners believe they can succeed and find a subject meaningful or relevant, their motivation is likely to be higher, leading to increased effort and better performance.

The knowledge of the key differences between expectancy theory and expectancy-value theory is significant. Expectancy theory focuses on the belief that efforts will lead to performance and performance will result in desired outcomes. In contrast, expectancy-value theory incorporates not only the expectation of success but also emphasizes the perceived value or importance of the outcome, suggesting that motivation is influenced by both expectancy and the subjective value attached

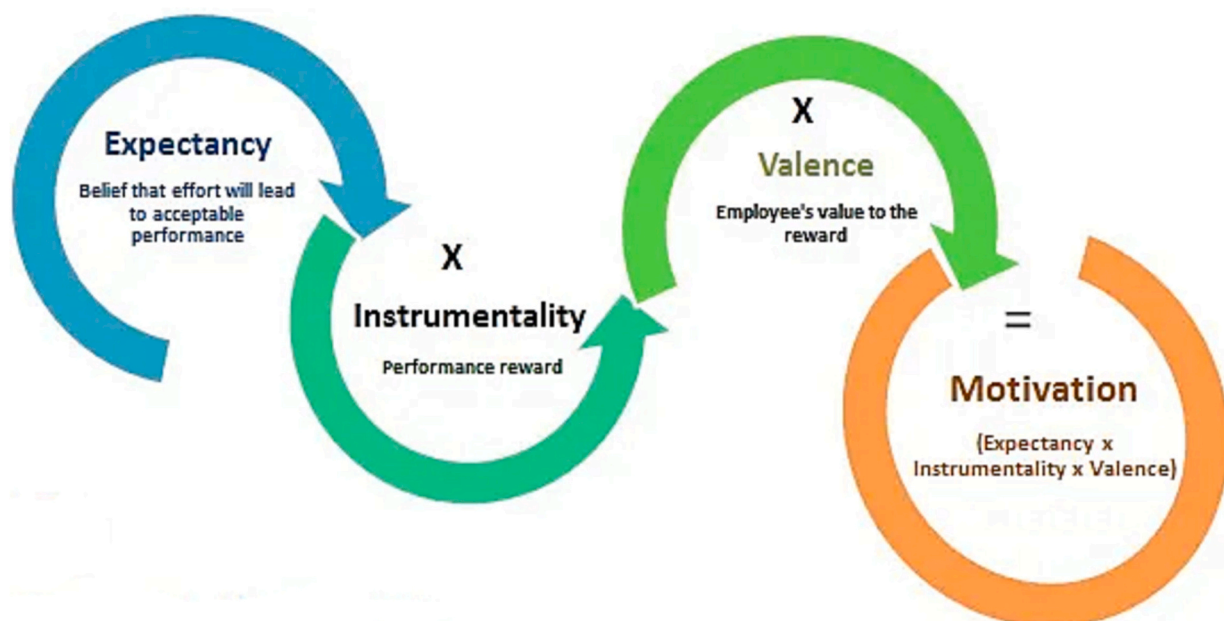


Fig. 3. Vroom's expectancy theory (Parijat & Bagga, 2014).

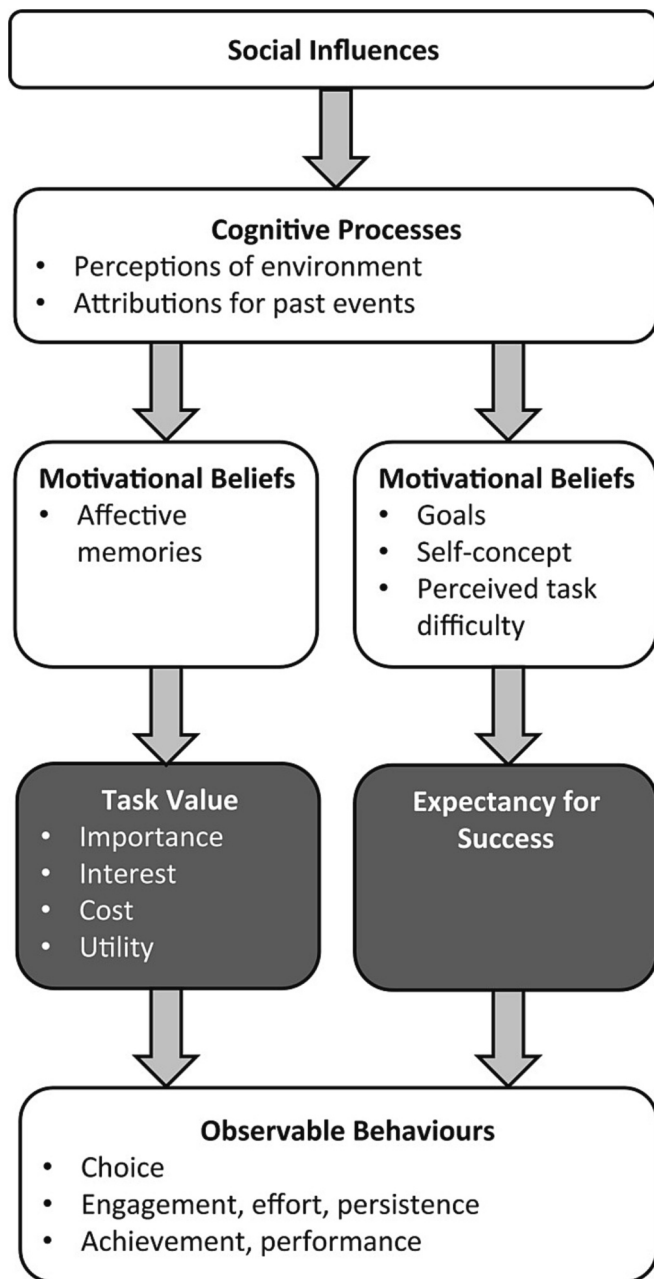


Fig. 4. Wigfield and Eccles's theory.

to achieving a goal (Estes & Polnick, 2012; Filipova, 2016).

12. ARCS model of motivation

The ARCS Model of Motivation is a framework for designing and evaluating instructional materials, which stands for Attention, Relevance, Confidence, and Satisfaction. The model was developed by John Keller in the 1980s and is based on the idea that motivation is a key factor in learning and that instructional materials should be designed to meet the motivational needs of learners (Keller, 1987). The ARCS model consists of four components as shown in Fig. 5 (Park, 2018).

- **Attention:** The instructional materials must capture the attention of the learners by being interesting, novel, and relevant to their needs and interests.

- **Relevance:** The materials must be relevant to the learners' needs, interests, and goals, and must be presented in a way that connects to the learners' previous knowledge and experiences.
- **Confidence:** The materials must provide the learners with a sense of confidence and self-efficacy by providing clear instructions, the appropriate level of difficulty, and feedback.
- **Satisfaction:** The materials must provide the learners with a sense of satisfaction and achievement by providing opportunities for success, problem-solving, and creativity.

The model is considered as a problem-solving tool that can help educators and designers to assess the motivational appeal of the materials they are using or planning to use and make improvements if necessary.

The ARCS Model is widely used in education, training, and instructional design, and has been used to design instructional materials for a wide range of learners, including young learners, adults, and special needs populations. The model is considered a useful framework for designing instruction that is motivating and effective, but it's important to note that it's not the only tool that can be used to design instruction, and other factors such as cognitive load, and cultural considerations, should also be taken into account (Keller, 2000; Li & Keller, 2018).

In education, the ARCS Model has been applied by designing engaging lessons that capture learners' Attention, such as incorporating multimedia or real-world examples. Relevance is achieved by connecting the material to learners' interests or future goals, fostering intrinsic motivation. Building confidence involves providing scaffolding and positive feedback, while satisfaction is addressed by creating meaningful assessments, ensuring learners feel a sense of accomplishment. In training, ARCS is utilized similarly, tailoring content to meet the needs and interests of trainees to enhance motivation and learning outcomes. In instructional design, the model guides the development of courses that are not only informative but also captivating, ensuring learners are attentive, perceive the material as relevant, feel confident in their abilities, and ultimately find satisfaction in their learning experience. Critics argue that the ARCS model oversimplifies motivation by focusing on discrete components and may not address the complexity of individual motivations (Chang & Chen, 2015; Keller, 2010).

13. Self-determination theory (SDT) of motivation

The self-determination theory (SDT) of motivation is a psychological theory that explains how human behavior is influenced by the degree to which the behavior is autonomous or self-determined. The theory proposes that behavior that is autonomous or self-determined is more likely to be sustained over time and to lead to greater well-being and personal growth than behavior that is controlled by external factors or the individual's perceived lack of choice (Ryan, 2009).

SDT was developed by Deci and Ryan in the 1980s as an extension of Deci's earlier work on intrinsic motivation. The simplified version of SDT, consisting of its three major elements namely intrinsic, extrinsic, and amotivation has been illustrated in Fig. 6.

It suggests that people have three innate psychological needs: autonomy, competence, and relatedness, that must be met in order to maintain optimal motivation. According to SDT, the level of autonomy in behavior is influenced by the degree to which the behavior is in line with an individual's values, goals, and needs (Gagné, & E.D.-J. of O. behavior, undefined, 2005). The theory distinguishes between different types of motivation namely autonomous motivation, controlled motivation, and amotivation. *Autonomous motivation* refers to behavior that is in line with an individual's values, goals, and needs and is perceived as self-determined. *Controlled motivation* refers to behavior that is performed because of external rewards or to avoid punishment. *Amotivation* refers to the absence of motivation. It happens when an individual does not have a reason or belief that behavior or activity will lead to the desired outcome or goal (Deci & Ryan, 2008; Vallerand et al., 2008).

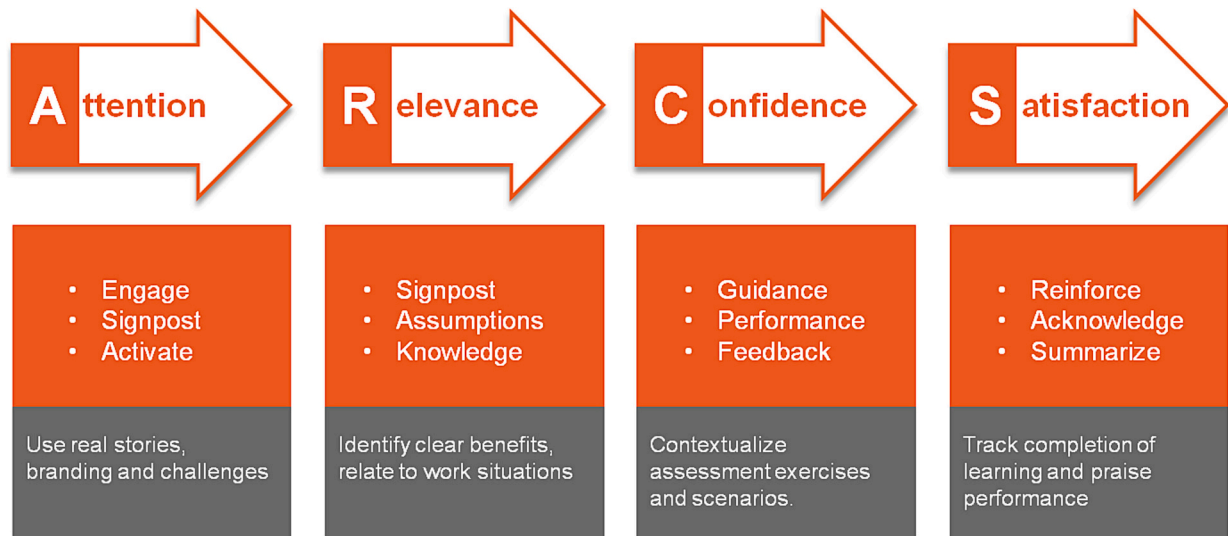


Fig. 5. Keller's ARCS model of motivation.

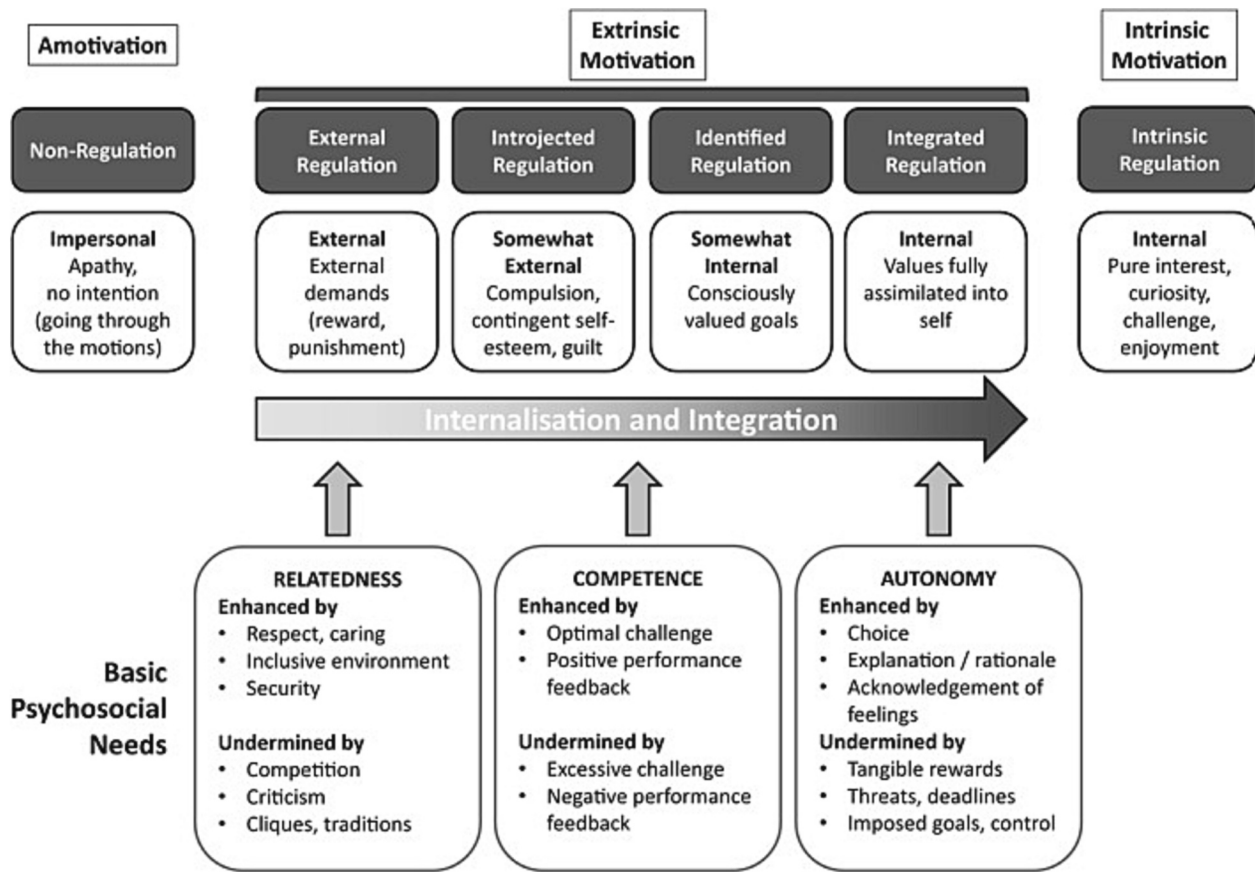


Fig. 6. Deci's self-determination theory.

The theory also suggests that when people feel autonomous and competent in their behavior, they are more likely to experience positive outcomes such as personal growth, well-being, and optimal performance. SDT has been applied in a wide range of fields, including education, sport, health, and the workplace, and has been used to explain the effects of different types of motivation on behavior and well-being. Research suggests that promoting autonomy-supportive environments and behaviors can lead to better outcomes, such as improved performance and satisfaction (Sheldon et al., 2014; Teixeira et al., 2012).

The traditional debates between learning theory, with a focus on extrinsic motivation, and SDT, which emphasizes intrinsic motivation and autonomy, have been central in understanding motivational processes and their implications for human behavior and achievement. Learning theory, rooted in behaviorism, highlights the role of external rewards and reinforcements in shaping behavior. It posits that individuals are motivated to engage in activities to obtain desired outcomes or avoid punishment. This perspective emphasizes the importance of extrinsic motivators, such as grades, rewards, and praise,

in driving behavior and learning. Despite this, it has been criticized for its limited consideration of internal motivational factors and the neglect of individuals' innate psychological needs. On the other hand, SDT emphasizes the intrinsic motivation that arises from individuals' inherent psychological needs for autonomy, competence, and relatedness. According to SDT, individuals are motivated when their activities align with their values, interests, and sense of personal autonomy. SDT recognizes the importance of internal motivators, such as enjoyment, curiosity, and self-determined choice, in promoting engagement, satisfaction, and well-being (Deci & Ryan, 2008; Sheldon et al., 2014; Teixeira et al., 2012).

In comparing these two perspectives, it becomes apparent that learning theory focuses primarily on external factors, while SDT emphasizes the interplay between internal and external motivational factors. While learning theory may be effective in certain contexts, it often overlooks the role of autonomy and intrinsic motivation, which have been found to be crucial for long-term engagement and optimal performance. SDT provides a broader and more comprehensive conceptualization of motivation by considering individuals' psychological needs and the quality of motivation. It posits that intrinsic motivation, which stems from the satisfaction of psychological needs, leads to greater engagement, creativity, and well-being. In contrast, extrinsic motivation can vary in its degree of autonomy, ranging from controlled to autonomous forms. SDT highlights the importance of fostering autonomous forms of extrinsic motivation that align with individuals' values and interests (Reeve, n.d.; Grolnick & Ryan, 1987; Vansteenkiste et al., 2006).

The interplay between learning theory and SDT has been a subject of empirical investigation. Studies have examined the complex dynamics and potential integration of these two motivational factors. Research has shown that external rewards can undermine intrinsic motivation when they are perceived as controlling or used as a means of exerting control. Regardless, when external rewards are provided in a way that supports individuals' autonomy and competence, they can enhance intrinsic motivation and foster a sense of ownership and engagement. Furthermore, studies have explored how the learning environment can be structured to support intrinsic motivation and autonomy. For example, providing meaningful choices, promoting competence-building experiences, and nurturing supportive relationships have been found to enhance intrinsic motivation and the quality of engagement. By bridging the gap between the traditional debates, the literature demonstrates the relevance and applicability of both learning theory and SDT in real-world contexts. It highlights the need for a balanced approach that considers both extrinsic and intrinsic motivators and acknowledges the importance of autonomy, competence, and relatedness in promoting optimal motivation and well-being (Grolnick & Ryan, 1987; Niemiec & Ryan, 2009; Vansteenkiste et al., 2006).

14. Social cognitive theory (SCT) of motivation

Social cognitive theory (SCT) of motivation is a psychological theory that suggests that behavior is influenced by the interaction between personal, behavioral, and environmental factors. SCT theory of motivation was developed by Albert Bandura in the 1980s. It is an extension of Bandura's earlier work on social learning theory (SLT), which proposed that individuals learn from observing and imitating the behaviors of others. SCT suggests that motivation is influenced by the interaction between personal, behavioral, and environmental factors (Bandura, 2005). According to SLT, people learn not only through direct reinforcement but also by observing the behaviors of others and the consequences of those behaviors. This observation can lead to the acquisition of new behaviors, as well as the shaping of existing behaviors. One of the key concepts in SLT is observational learning, which is the process of learning new behaviors by observing others. This process involves four main steps: attention, retention, reproduction, and motivation. Bandura conducted several experiments to demonstrate the power of

observational learning. In one of his most famous experiments, the Bobo doll experiment, Bandura showed that children were more likely to imitate aggressive behavior after watching an adult model behave aggressively. Fig. 7 depicts the modified version of Schunk's model of motivated learning. It incorporates additional concepts from Bandura and other authors.

Schunk's model of motivated learning provides a framework for understanding the factors that influence learners' motivation to learn. The model incorporates several key concepts from social cognitive theory (SCT), emphasizing the interplay between personal, behavioral, and environmental factors in shaping motivation. Key alignments between Schunk's Model and SCT include self-efficacy, observational learning, goal-setting, and self-regulation.

SCT expands upon previous theories by integrating cognitive, behavioral, and social aspects of motivation. Unlike earlier theories that focused primarily on either internal drives or external rewards, the social cognitive theory emphasizes the interaction between individuals and their environment. One key aspect that sets social cognitive theory apart is its emphasis on self-efficacy beliefs. Self-efficacy refers to an individual's belief in their ability to successfully perform a specific task or behavior. This concept allows for a more nuanced understanding of motivation as it recognizes the influence of one's perceived competence on their motivation to engage in specific activities. SCT also highlights the role of observational learning, where individuals acquire new behaviors and motivations by observing others. This expands the scope of motivational processes beyond individual experiences and includes the influence of social modeling and vicarious reinforcement (The Social Cognitive Theory, n.d.; Conti & Amabile, 2011; Stavropoulou et al., 2023; Thomson & Jaque, 2017).

According to SCT, motivation is driven by the individual's goals and expectations and is influenced by the perceived outcomes of the behavior, the individual's self-efficacy (i.e., the belief in one's ability to perform the behavior), and the individual's perceived control over the situation. The theory also suggests that people are more likely to engage in behavior that they perceive as leading to desired outcomes when they: i). have a high level of self-efficacy, ii). perceive that the behavior is under their control, and iii). perceive that the behavior is consistent with their goals and values (Bandura, 2002; Stavropoulou et al., 2023; Wood & Bandura, 1989).

Each learner enters a learning encounter with a unique sense of confidence based on their unique blend of prior knowledge, natural abilities, and social reinforcements. A learner can either do the activity by themselves or observe another learner (either an instructor or a peer model) doing the activity. Cognitive involvement, motivation to learn, emotional reaction, and work evaluation are all affected by self-efficacy in addition to other personal and environmental aspects during the activity. Self-efficacy for subsequent tasks is influenced by indicators that learners pick up, both during and after work. Like this paradigm, Zimmerman's self-regulation cycle consists of three stages: pre-task planning (pre-task), task execution with self-directed effort (during the task), and post-task evaluation (after the task).

The SCT offers several comparative strengths that make it a valuable framework for understanding motivation, surpassing some of the limitations of previous theories. By addressing these strengths, it can be highlighted why social cognitive theory provides a better explanation of motivation. Schunk's model specifically incorporates concepts from Bandura's work on SCT as follows (Alahmad, 2020; Toomey & Heo, 2022; Urhahne & Wijnia, 2023; Young et al., 2014):

- *Integration of Cognitive and Behavioral Factors:* Unlike earlier theories that predominantly focused on either internal drives or external rewards, SCT integrates both cognitive and behavioral factors. It recognizes that motivation is influenced not only by external stimuli but also by internal cognitive processes, such as beliefs, expectations, and self-perceptions. This integration allows for a more comprehensive understanding of motivation.

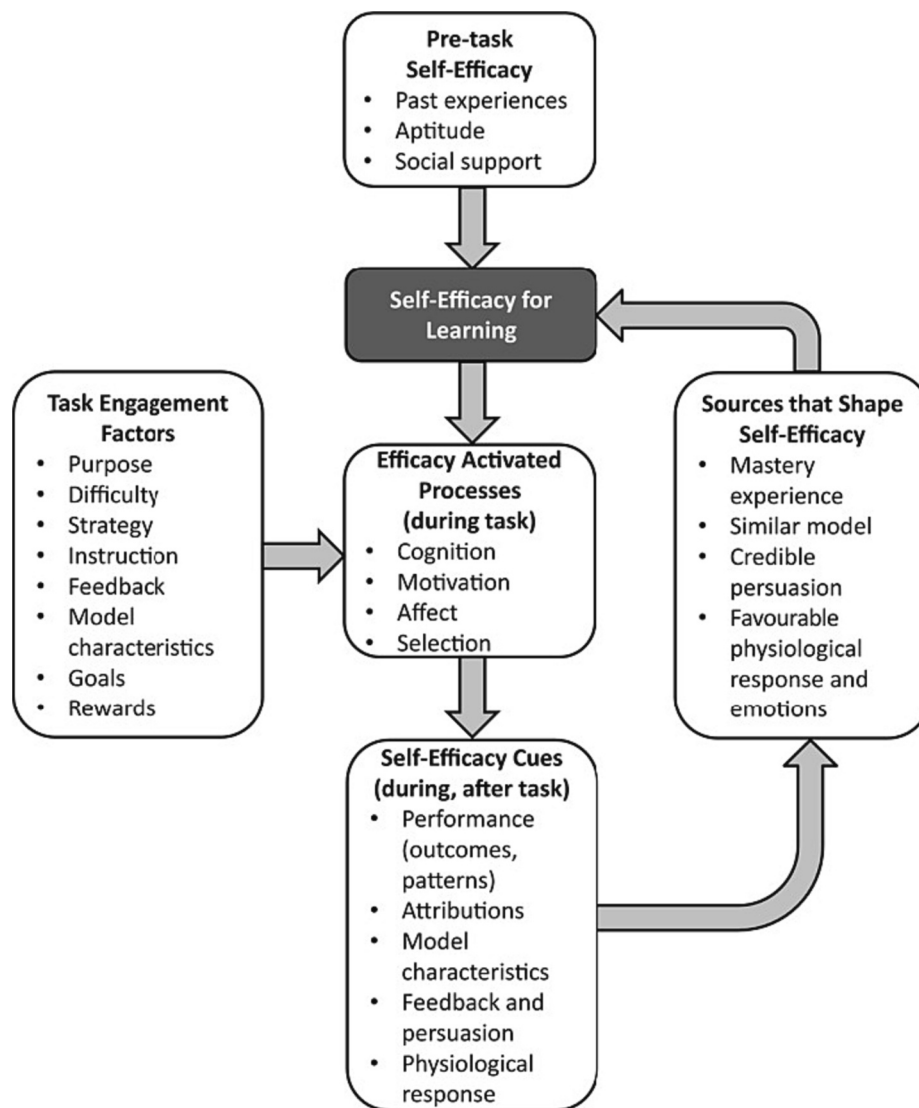


Fig. 7. Schunk's model of motivated learning.

- **Emphasis on Self-Efficacy:** One of the key strengths of SCT is its emphasis on self-efficacy beliefs. Self-efficacy refers to an individual's belief in their ability to successfully perform a specific task or behavior. This concept addresses the question of "Can I do it?" and plays a crucial role in motivating individuals to engage in activities. Unlike previous theories that may have overlooked the role of self-beliefs, SCT recognizes the significance of self-efficacy in shaping motivation.
- **Observational Learning and Vicarious Reinforcement:** SCT acknowledges the influence of observational learning and vicarious reinforcement on motivation. Individuals can acquire new behaviors and motivations by observing others and witnessing the outcomes of their actions. This expands the scope of motivational processes beyond individual experiences and highlights the social aspects of motivation. By considering the role of observational learning, SCT provides a richer understanding of how individuals' motivations are influenced by their social environment.
- **Contextual Factors and Reciprocal Determinism:** SCT recognizes the importance of contextual factors and the dynamic interplay between individuals and their environment. It proposes the concept of reciprocal determinism, suggesting that individuals and their environment mutually influence each other. This perspective acknowledges that motivation is not solely an individual phenomenon but is

shaped by social and environmental factors. By considering the reciprocal relationship between individuals and their context, SCT offers a more comprehensive explanation of motivation.

When comparing SCT to previous theories, its strengths become apparent. For instance, compared to intrinsic and extrinsic theories that focus primarily on internal instincts needs, or incentives, SCT provides a more comprehensive understanding by considering cognitive processes, observational learning, and the influence of the social context. Additionally, SCT's emphasis on self-efficacy distinguishes it from expectancy-value theories that may have provided a broader, less task-specific view of beliefs about competence. By incorporating these comparative strengths, it can be demonstrated how SCT offers a more nuanced and comprehensive explanation of motivation compared to previous theories. The integration of cognitive and behavioral factors, emphasis on self-efficacy, recognition of observational learning, and consideration of contextual factors contribute to a more robust understanding of motivation (Young et al., 2014).

SCT has been applied in a wide range of fields, including education, health, sport, and the workplace. For example, In the classroom, teachers can foster both intrinsic and extrinsic motivation by providing stimulating learning activities, allowing learners to pursue their interests, and offering clear goals and expectations. However, overemphasis on

extrinsic rewards can undermine intrinsic motivation, so it's crucial to find a balance between the two. Also, in sports, coaches can enhance athletes' motivation by fostering self-efficacy through positive reinforcement, providing realistic goals, and breaking down tasks into achievable steps. By emphasizing self-belief, coaches can empower athletes to pursue challenging goals and persist in the face of setbacks. It has been used to explain how people learn from observing and imitating the behaviors of others and how this learning influences their behavior. It also has been used to explain how people are motivated to engage in behaviors that they perceive as leading to desired outcomes, and how factors such as self-efficacy, perceived control, and goal-setting can influence motivation (Bandura, 1988; Ilmiani et al., 2021).

15. Goal-orientation theory of motivation

The goal-orientation theory of motivation is a psychological theory that suggests that individuals differ in their goals or reasons for engaging in a behavior or activity. The theory proposes that individuals can be classified into different goal orientations, such as a performance-approach goal orientation, a performance-avoidance goal orientation, a mastery-approach goal orientation, and a work-avoidance goal orientation. Performance-approach goal orientation refers to the tendency to engage in a task to demonstrate one's ability and outperform others. Performance-avoidance goal orientation refers to the tendency to engage in a task to avoid failure and negative evaluation by others. Mastery-approach goal orientation refers to the tendency to engage in a task to improve one's skills, knowledge, and abilities. Work-avoidance goal orientation refers to the tendency to avoid engaging in a task

because it's viewed as unimportant or uninteresting (Domurath et al., 2020; Lam et al., 2022).

The theory was first proposed by Dweck and Leggett in 1988 and later by Elliot and McGregor in 2001, and it has been applied in various fields such as education, sport, and the workplace. The simplified version of Dweck's theory has been illustrated in Fig. 8.

Carol Dweck and Ellen Leggett identified two distinct goal orientations: performance-goal orientation and mastery-goal orientation. Performance-goal orientation is characterized by a focus on demonstrating ability and achieving external validation, often through comparisons with others. Individuals with a performance-goal orientation are primarily motivated by the desire to outperform others, avoid negative judgments, and gain social approval. Mastery-goal orientation, on the other hand, emphasizes learning, self-improvement, and personal growth. Individuals with a mastery-goal orientation are intrinsically motivated by the desire to enhance their skills, understand concepts, and overcome challenges. They view setbacks as opportunities for learning and growth rather than threats to their self-esteem (Reeve, n.d.; Hattie et al., 2020).

In 2002, Andrew Elliot and Helen McGregor expanded upon Dweck and Leggett's theory, proposing a more complex and multifaceted framework for understanding goal orientations. They identified two dimensions of goal orientation: task-goal orientation and ego-goal orientation. Task-goal orientation is similar to mastery-goal orientation, emphasizing learning, self-improvement, and the development of competence. Individuals with a task-goal orientation are motivated by the inherent satisfaction of mastering new skills and solving problems. Ego-goal orientation is similar to performance-goal orientation, focusing

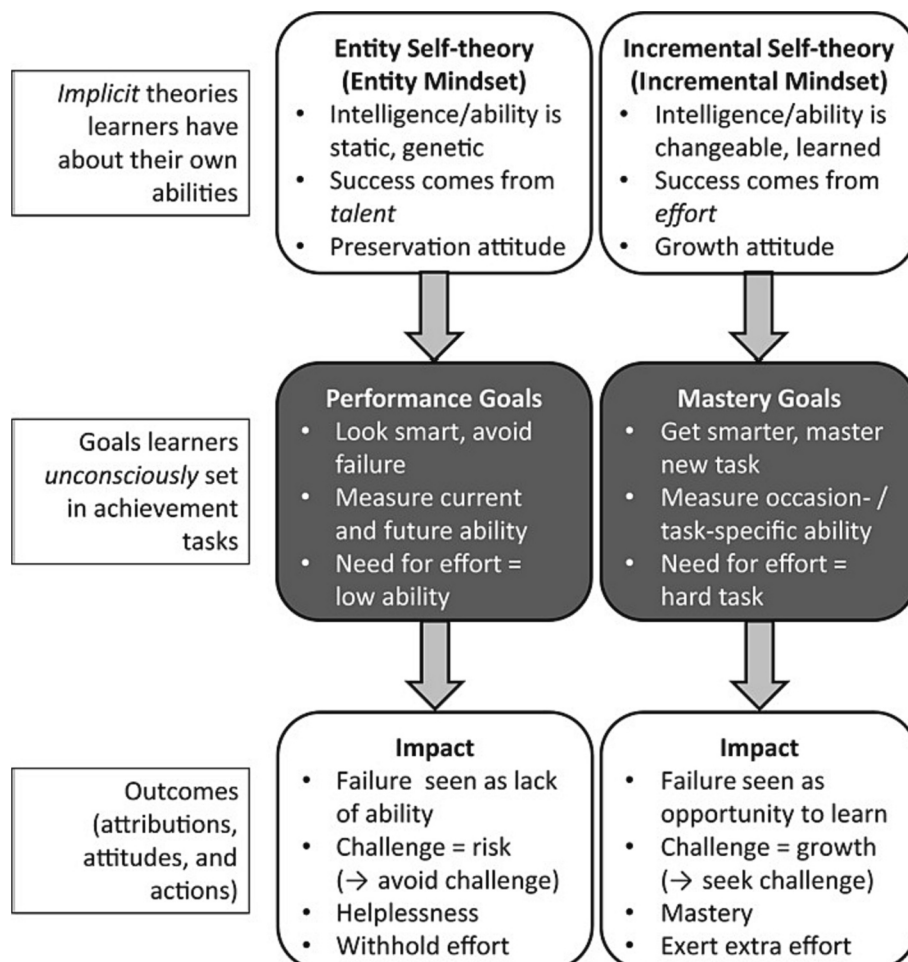


Fig. 8. Dweck's goal-orientation theory.

on demonstrating ability and achieving external validation. However, Elliot and McGregor further differentiated ego-goal orientation into two subtypes. The first one is performance-approach ego-goal orientation. This subtype is characterized by a focus on outperforming others and achieving superior performance. Individuals with this orientation are motivated by the desire to gain recognition, status, and social approval. The second one is performance-avoidance ego-goal orientation. This subtype is characterized by a fear of failure and a desire to avoid negative judgments from others. Individuals with this orientation are motivated by the need to avoid embarrassment, criticism, and damage to their self-esteem (Linnenbrink & Pintrich, 2002; Wang et al., 2021).

While both Dweck and Leggett's and Elliot and McGregor's interpretations of goal-orientation theory share the core distinction between performance and mastery goals, they also differ in their complexity and nuance. Both models emphasize the significant impact of goal orientation on individuals' motivation, persistence, and achievement. Both models recognize the importance of both performance and mastery goals, acknowledging that individuals may exhibit both orientations to varying degrees. Elliot and McGregor's model provides a more refined and multifaceted framework, differentiating ego-goal orientation into two subtypes. Elliot and McGregor's model places greater emphasis on the motivational consequences of goal orientations, particularly the contrasting effects of task-goal orientation and performance-approach versus performance-avoidance ego-goal orientation (McCullum & Kajs, 2007; Wang et al., 2021).

The goal-orientation theory of motivation has evolved from Dweck and Leggett's original proposition to Elliot and McGregor's more complex interpretation. Both models provide valuable insights into how individuals' goal orientations influence their motivation, persistence, and achievement. Educators, coaches, and individuals alike can benefit from understanding the dynamics of goal orientations and fostering a focus on mastery-oriented and task-goal-oriented goals to promote intrinsic motivation, learning, and personal growth. Research suggests that individuals with a mastery-approach goal orientation tend to perform better, experience more satisfaction, and have a better well-being than those with a performance-approach or performance-avoidance goal orientation. The theory has been used to design instruction, improve motivation, and increase well-being and performance. For example, teachers can foster a mastery-approach goal orientation by designing tasks that are challenging but achievable, providing opportunities for personal growth, and giving feedback that emphasizes learning and progress over performance (Elliot et al., 2011; Urdan & Kaplan, 2020).

16. Mutual themes

Drawing from the preceding discussions, four main themes emerge from these theories that can be used to better understand the relationships and differences between them. Accordingly, a concise overview of each theory is provided in Table 1.

- **Competence:** Perceptions of one's *competence* are deeply embedded within the framework of these motivational theories. Expectancy of success, self-efficacy, confidence, and self-concept are all names for the same set of ideas that aim to answer the question "Can I do it?" There are, nevertheless, significant differences between and within theories, as previously discussed. Self-concept and older concepts of anticipation of success (expectancy-value theory) saw such views in broad strokes (e.g., covering a large area like 'athletics' or 'clinical medicine,' or broadening across time or circumstances). On the other hand, self-efficacy (social-cognitive theory) and subsequent concepts of expectation of success interpreted similar views in a variety of task- and situation-specific contexts (e.g., "Can I assess the seriousness of cardiac stenosis?").
- **Value:** The *value* of the learning activity, or its expected outcome, is also typically discussed in these theories. Beliefs like task worth,

expected outcome, and internal vs. external motivation are examples of such concepts. All of these answers deal with the dilemma of "Do I want to do it?" or "What will occur (positive or negative) if I do it?" Once again, there are substantial gaps between competing theories. For instance, task value (expectancy-value theory) emphasizes the perceived significance or utility of successfully completing a task, while outcome expectancy (social-cognitive theory) emphasizes the likely (anticipated) consequence of action if maximum commitment is spent.

- **Attributions:** Most theories highlight the role of *attributions* in influencing attitudes and behavior. Whether consciously or unconsciously, learners often attribute external events or outcomes to internal elements that contributed to them. Learners are more inclined to keep trying after an early setback if they believe the fundamental cause is malleable and under their control.
- **Cognitive:** Finally, all current motivation theories are 'cognitive' in the sense that, contrary to certain older theories, they all assume the participation of non-observable mental processes. Furthermore, contemporary theories have come to acknowledge that motivation is seldom an isolated individual phenomenon, but rather the result of intricate relationships between the individual and their social environment. Even though Bandura called his theory a "social-cognitive theory of learning," the other theories, discussed above, also incorporate both social and cognitive aspects of the learning process.

17. Summary and recommendation

This study has covered a number of motivation theories used in academic learning. Historically, self-determination theory (SDT) and the ARCS model were broadly applied in the motivation purview for learning activities. Theoretical frameworks like the social cognitive theory and the expectancy theory are just starting to be put into practice. The goal throughout was to identify models with the potential to boost learner engagement in the field of education. Learning is a complex activity that requires direction if it is to be accomplished effectively. Theories are crucial to the success of determined endeavors. A diversion from the necessary goal would be disastrous. As a result, scholars have developed theoretical frameworks to direct these activities. Theoretical frameworks are useful for organizing and executing educational activities, and they also offer pointers for dealing with real-world difficulties.

Similar to other psychological theories, motivation structures have led to several competing hypotheses that endeavor to describe comparable behaviors. The question of whether motivation is a function, a trait, or a condition is itself a matter of debate. This conflicting yet interconnected nature of these theories makes it harder to get accurate information while researching a phenomenon academically. Everyone on a phenomenological level knows that intrinsic motivation is a key factor in learning, but traditional methods of measuring rely on self-report (which is notoriously incorrect) or on observable changes in behavior (which involves circular reasoning). This predicament raises a number of intriguing concerns. The first is the contrast between the ideas of contextual motivation and self-regulation of motivation. The former view, which is congruent with behavioral psychology, has only lately been considered in the research of learning goal concepts and their influence on academic motivation, holds that motivation is driven by ecological variables. When a learner engages in self-regulated motivation, he or she uses active regulatory tactics to manage his or her motivation. Different suggestions for research and improvement of motivation are provided by these two theories. A common theme across contemporary theories, namely social cognitive theory, is the idea that one's degree of motivation is contingent on a dynamic exchange between the individual and their surrounding ecosystem.

The connection between intrinsic and extrinsic forces/factors of motivation offers a rich vein of intriguing unanswered issues. When compared to motivation that comes from inside or is intrinsically valuable to the endeavor, what is the value of influence that an extrinsic form

Table 1
Overview of theories of motivation.

	Main Idea	Key Concepts	Key Differences
Extrinsic theory	Individuals are driven by external rewards, such as money, grades, or recognition, to engage in certain behaviors.	The focus is on the external factors that influence motivation, such as incentives, punishments, and social recognition. It assumes that people's behavior is primarily driven by external factors rather than internal desires or interests.	In contrast to intrinsic theory, which emphasizes internal motivation and personal satisfaction, extrinsic theory highlights the role of external rewards in shaping behavior.
Attribution theory	- After an incident, learners unconsciously generate causal explanations (attributions) for the outcomes. - Attributions differ from the viewpoint of locus, stability, and controllability. These have an impact on feelings, which in turn motivates future activities.	- Locus indicates whether an individual's problem has an internal or external source. - Stability denotes whether the underlying cause is unchanging or subject to change. - Controllability indicates whether or not the underlying reason is under the control of the person.	- Attribution theory distinguishes from expectancy-value theory by the premise that emotional mediation of task value. - The goal-orientation theory relies heavily on implicit attributions regarding the consistency of one's abilities. - According to social-cognitive theory, attributions play a crucial role in the process of self-regulation.
Expectancy theory	Learners are motivated when they believe their efforts will lead to desired performance and outcomes.	The theory involves three key components: expectancy (the belief that effort will result in performance), instrumentality (the belief that performance will lead to desired outcomes), and valence (the value or importance assigned to the desired outcomes).	Expectancy theory is distinct from other motivation theories in its focus on the cognitive processes underlying motivation and the belief in the link between effort, performance, and outcomes. It emphasizes individuals' perceptions of the likelihood and desirability of outcomes.
Intrinsic theory	Internal factors, such as personal interests, enjoyment, and the inherent satisfaction derived from engaging in an activity play a significant role in driving human behavior.	- The theory emphasizes the intrinsic rewards associated with an activity, such as the sense of accomplishment, enjoyment, or personal growth. - It suggests that individuals have an innate drive to pursue activities that align with their interests and provide intrinsic satisfaction.	The intrinsic theory stands in contrast to the extrinsic theory, as it highlights the role of internal factors and personal satisfaction rather than external rewards in driving motivation.
Expectancy-value theory	Motivation is influenced by expectations of achievement and perceived worth.	- Success expectation is the confidence that one will achieve their goal if one puts out the effort. - A person's job value is determined by how much they value or enjoy performing the activity for themselves.	Several other theories also make use of ideas like an expectancy of success and value.
ARCS model of motivation	The ARCS (Attention, Relevance, Confidence, and Satisfaction) Model of Motivation focuses on designing instructional materials and strategies to enhance learner motivation.	The model emphasizes creating a learning environment that captures learners' attention, demonstrates the relevance of the material to their goals and interests, builds their confidence in their ability to succeed, and provides a sense of satisfaction or accomplishment.	The ARCS Model complements other motivation theories by providing practical guidelines for instructional design and implementation, taking into account specific motivational factors that can enhance engagement and learning outcomes.
Self-determination theory (SDT)	- SDT encompasses both intrinsic and extrinsic motivations contextualized by these three factors: - Intrinsic motivation makes people do things just to satiate their inquisitiveness or urge to get better at something. - Social values are what drive extrinsic motivation, which is what drives all other actions. Integrating and internalizing one's values can transform behavior that was once motivated by external factors into a self-determined act. - Feelings of competence, autonomy, and relatedness are conducive to intrinsic and internalized motivations.	- Autonomy is the freedom to make one's own decisions. - Competence describes a person's apparent skill and success. - Relatedness is a feeling of connection or being associated with other people with whom you'd like to feel linked.	Self-determination theory is distinguished from other theories by its focus on autonomy, choice, and human relationships.
Social cognitive theory (SCT)	- Personal characteristics, behavioral tendencies, and external context all interact to influence an individual's ability to learn and perform. - Self-efficacy is the main source of motivation.	- Self-efficacy is a personal assessment of one's capacity for learning or performance at a certain level. - Outcome expectation describes the view that certain outcomes will come from certain acts. - Self-regulation entails a cyclical pattern of goal pursuit, wherein individuals monitor their progress and adjust their strategies based on self-assessment.	Though it shares some similarities with other constructs of expectancy, self-efficacy tends to be more task-, context-, and goal-oriented.
Goal-orientation theory	- Learners often look at activities with three main objectives in mind: comprehending the content (the mastery objective), outperforming their peers (the performance approach objective), and preventing themselves from failure (the performance-avoidance objective). - While performance-based goals are linked to higher grades, mastery goals seem to promote curiosity and in-depth study. Negative results are more often related to performance-avoidance aims.	- Focusing on improving one's intelligence or skill is called a "mastery orientation," and it stems from a "growth learning mindset" (skills can be shaped and circumstances can be managed). - Performance orientation emphasizes seeming intelligent rather than stupid; it results from the learning mentality known as "entity" (ability is fixed and circumstances are less under one's control).	Contrary to attribution theory, controllability beliefs differ from person to person (i.e., they are not a constant attribute of the event or learner).

of motivation may have? Specifically relevant to teachers who seek to encourage self-directed, perpetual learning yet struggle to ignite their learners' enthusiasm in the conventional coursework and frequently turn to grades as a kind of motivation, problems like these demand solutions. Emotional influences on motivation go beyond this, it is an emerging field of study which highlights the brain's role in determining behavior. This segment began with difficulty, the requirement for a better accurate indicator of motivation, and a possible answer may lay in the capacity to assess the brain's responsiveness to transformations in the ecosystem at multiple levels.

CRedit authorship contribution statement

Din Bandhu: Writing – review & editing, Writing – original draft, Visualization, Conceptualization. **M. Murali Mohan:** Writing – original draft, Visualization. **Noel Anurag Prashanth Nittala:** Writing – review & editing, Resources, Methodology. **Pravin Jadhav:** Supervision, Resources, Formal analysis. **Alok Bhadauria:** Writing – review & editing, Resources, Funding acquisition. **Kuldeep K. Saxena:** Visualization, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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